Фильтр пакетов

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13 марта 2026

Contents

Цели и задачи работы **2**
Цель лабораторной работы 2

Процесс выполнения лабораторной работы **3**
Определение конфигурации 3
Анализ активной зоны 4
Добавление сервиса (временное изменение) 5
Добавление сервиса(постоянное изменение) 6
Добавление порта 2022/TCP 7
Включение служб 8
Добавление порта 9
Проверка результата 10
Добавление служб 11
Проверка конфигурации 12

Выводы по проделанной работе **13**

Вывод 13

Цели и задачи работы

Цель лабораторной работы

Получить навыки работы с межсетевым экраном в Linux и научиться управлять правилами фильтрации пакетов через:

- firewall-cmd (CLI)
- firewall-config (GUI)

Процесс выполнения лабораторной работы

Определение конфигурации

-

```
root@localhost:/home/manieyvazi# firewall-cmd --get-default-zone
public
root@localhost:/home/manieyvazi# firewall-cmd --get-zones
block dmz drop external home internal nm-shared public trusted work
root@localhost:/home/manieyvazi# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apc
upsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin b
itcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civi
lization-iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 d
hcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-cl
ient etcd-server factorio finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freei
pa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap
imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana kl
ogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manage
r kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kube
let-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp m
anagesieve matrix mdns memcache minecraft minidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mys
ql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry o
penvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql
privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius r
adsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane se
ttlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroa
k-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming stellaris stronghold-crusader stun s
tuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-tls te
lnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsm vnc-server vrrp war
pinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-discove
ry-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gatewa
y zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
root@localhost:/home/manieyvazi# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
```

Рис. 1 — Список зон и служб

Анализ активной зоны

-.

```
root@localhost:/home/manieyvazi# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
```

Рис. 2 — Параметры зоны public

Добавление сервиса (временное изменение)

•

```
root@localhost:/home/manieyvazi# firewall-cmd --list-all --zone-public
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --zone-public
root@localhost:/home/manieyvazi# firewall-cmd --add-service=vnc-server
success
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi#
```

Рис. 3 — Добавление VNC и исчезновение после рестарта

Добавление сервиса(постоянное изменение)

-.

```
root@localhost:/home/manieyvazi# firewall-cmd --list-all --zone-public
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --zone-public
root@localhost:/home/manieyvazi# firewall-cmd --add-service=vnc-server
success
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi#
```

Рис. 4 — Добавление VNC permanent

Добавление порта 2022/TCP

-.

```
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi# firewall-cmd --reload
success
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi# firewall-cmd --add-port=2022/top --permanent
Error: INVALID_PROTOCOL: 'top' not in {'tcp'|'udp'|'sctp'|'dccp'}
root@localhost:/home/manieyvazi# firewall-cmd --add-port=2022/tcp --permanent
success
root@localhost:/home/manieyvazi# firewall-cmd --reload
success
```

Рис. 5 — Добавление порта 2022/tcp

Включение служб

-.

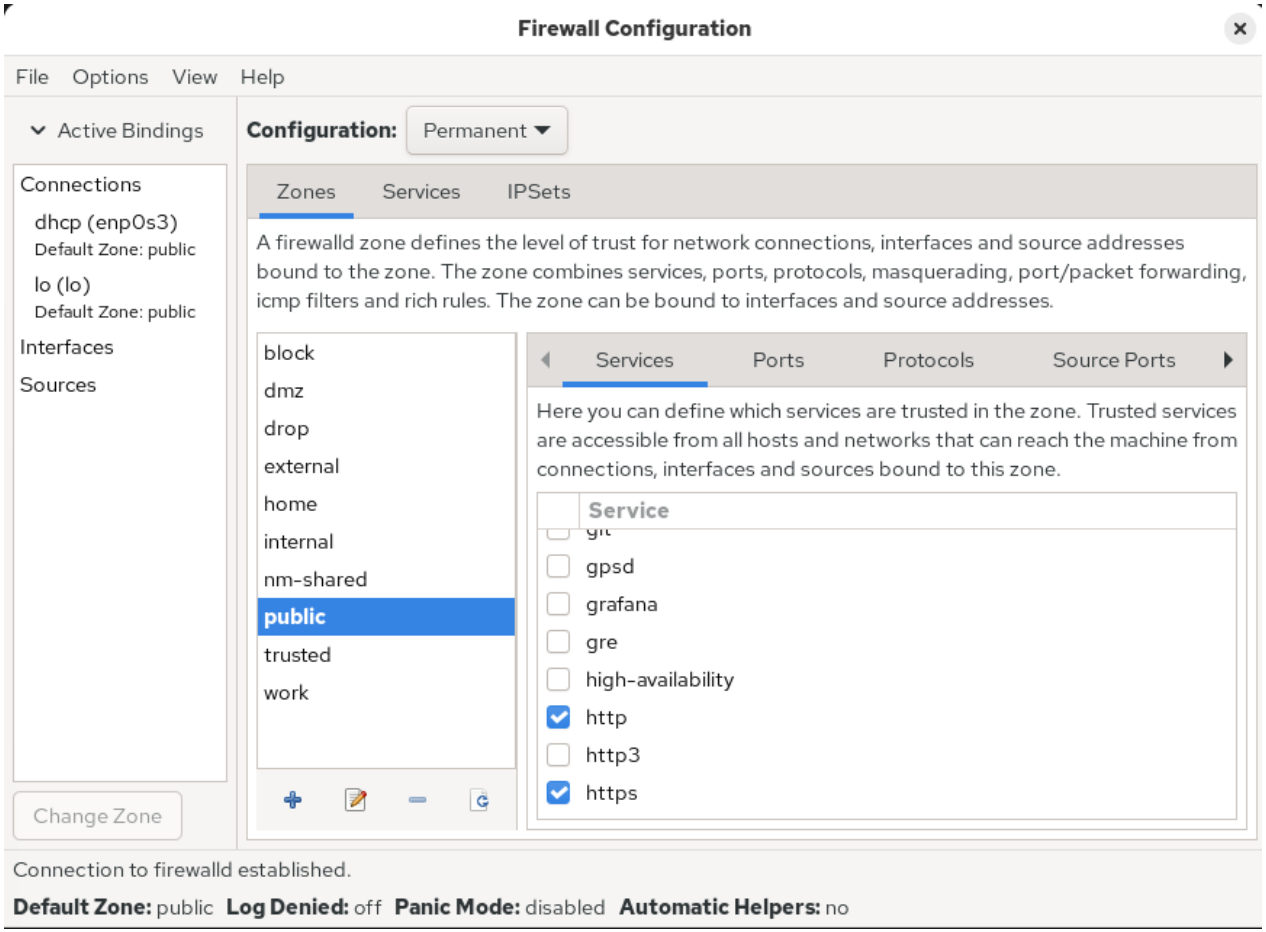

Рис. 6 — GUI firewall-config services

Добавление порта

-.

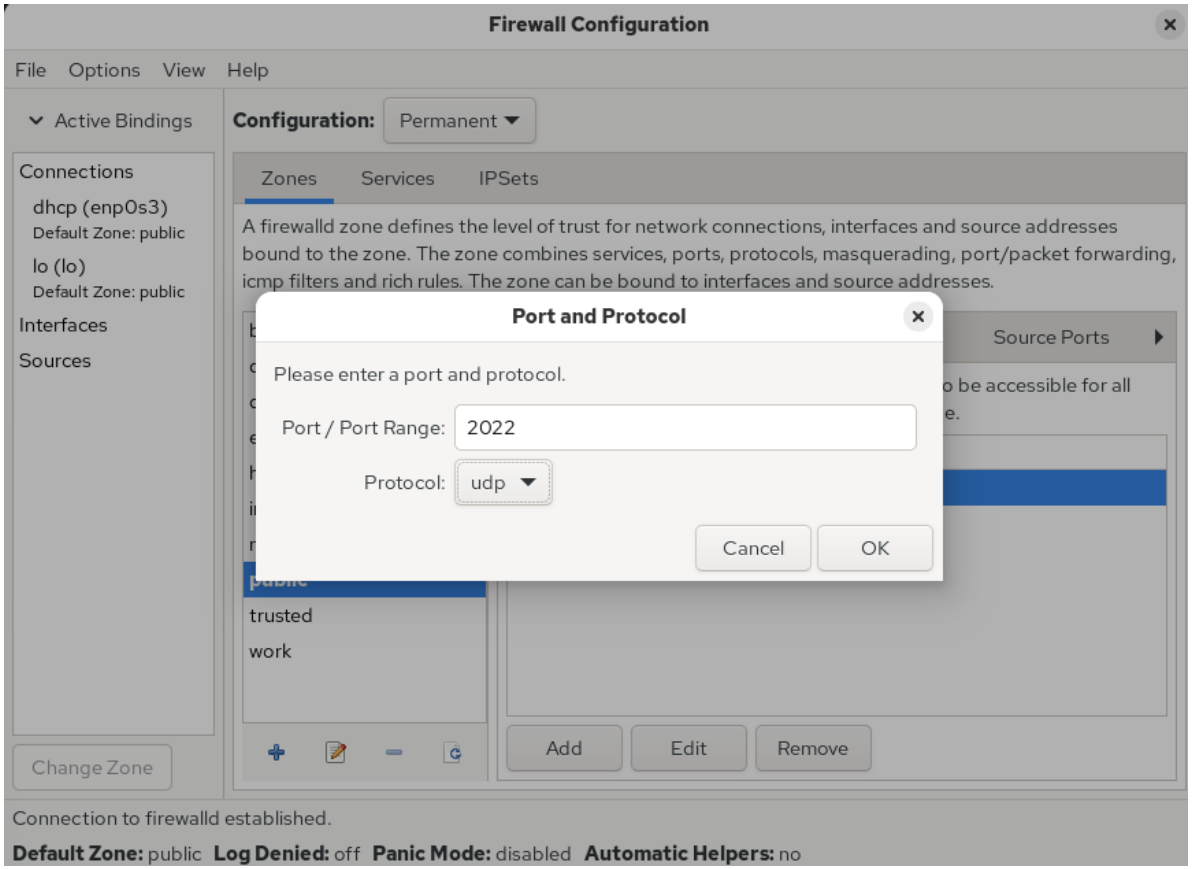

Рис. 7 — GUI firewall-config port

Проверка результата

-.

```
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports: 2022/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi# firewall-cmd --reload
success
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https ssh vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi#
```

Рис. 8 — Список активных служб и портов

Добавление служб

Проверка работы системы.

```
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports: 2022/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi# firewall-cmd --reload
success
root@localhost:/home/manieyvazi# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https ssh vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@localhost:/home/manieyvazi#
```

Рис. 9 — Добавление служб через GUI

Проверка конфигурации

-

Рис. 10 — Итоговая конфигурация

Выводы по проделанной работе

Вывод

- Изучены инструменты управления брандмауэром: **firewall-cmd** и **firewall-config**
- Освоены:
  - просмотр зон и разрешённых служб
  - добавление портов и сервисов
  - различие между временной и постоянной конфигурацией
- На практике закреплено применение параметра `--permanent` и активация изменений через `firewall-cmd --reload`